Rethinking Toric Code

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1 Toric code as a Gauge theory

The toric code is also referred to as a Z_2 gauge theory. To understand why it is a gauge theory, we can draw an analogy to electromagnetism, which is a well-known gauge theory due to its 'gauge symmetry'. Recall Maxwell's equations:

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{j} + \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t}. \tag{1}$$

The gauge symmetry in electromagnetism arises with the introduction of the electromagnetic potentials. The first equation is automatically satisfied if

$$\mathbf{B} = \nabla \times \mathbf{A}, \tag{2}$$

since $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for any $\mathbf{A}$. Substituting this into the second equation, we obtain

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0, \tag{3}$$

which is satisfied if

$$\mathbf{E} = -\nabla \phi - \frac{\partial \mathbf{A}}{\partial t}, \tag{4}$$

because $\nabla \times (\nabla \phi) = 0$ for any ϕ. The key observation is that given $\mathbf{E}$ and $\mathbf{B}$, the potentials $\mathbf{A}$ and ϕ are not uniquely determined. We get the same $\mathbf{E}$ and $\mathbf{B}$ if we transform $\mathbf{A}$ and ϕ as

$$\mathbf{A} \to \mathbf{A} + \nabla f, \quad \phi \to \phi - \frac{\partial f}{\partial t},$$

where f can be any real function of space and time. This is is the gauge transformation of the electromagnetic potentials, and $\mathbf{A}$ and ϕ are the gauge fields. Under this gauge transformation, the fields $\mathbf{E}$ and $\mathbf{B}$ remain invariant, meaning Maxwell's equations are invariant as well. Therefore, the gauge transformation represents a symmetry of Maxwell's equations. Unlike a global symmetry transformation, which is uniform across space and time, a gauge transformation can vary at each point in space-time, making it a local symmetry.

1.1 Lattice gauge transformation

Now, before going into TC, let us put the gauge theory of quantum electrodynamics on a lattice. In the quantized version of electromagnetism, classical observables become quantum operators. To define the quantum theory on a lattice, we introduce a link degree of freedom on each edge. The operator A represents the 'angular position', while E represents the 'angular momentum', as is

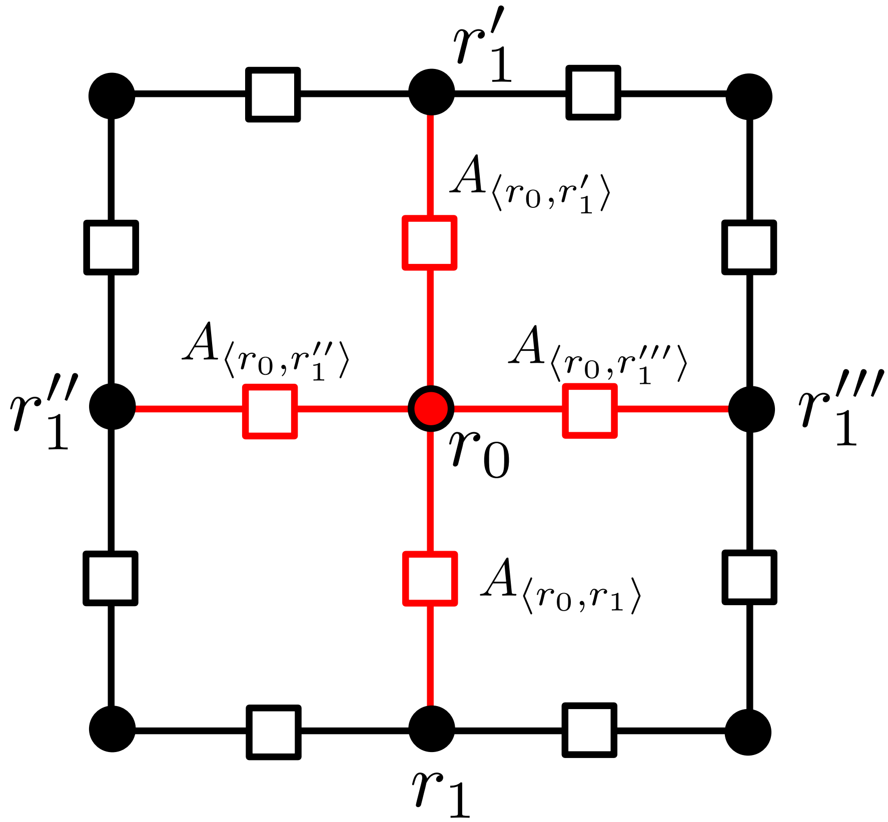

Figure 1: illustration of lattice dofs and gauge transformation. Link dofs (vector potential A) are represented by squares, while vertices are represented by circles. A local gauge transformation defined at site r_0 affects the four red links connected to it.

illustrated in Fig. 1. These operators form a canonical conjuage pair, satisfing the commutation relation

$$[A, E] = i. \tag{5}$$

On the lattice, we can partially fix the gauge by setting $\phi = 0$ and focusing on the spatial part of the gauge transformation:

$$A \to A + \nabla f. \tag{6}$$

For a compact U(1) theory, we define

$$A \in [0, 2\pi) \tag{7}$$

hence its conjuage E is restricted to

$$E \in \mathbb{Z} \tag{8}$$

since the eigenstate of E takes a plane waveform $\psi_n = e^{inA}$, and must satisfy $\psi_n = \psi_{n+2\pi}$.

Now we need to define a lattice version of the gauge transformation in Eq. (6). We would like to construct a local theory, hence we define the function f to be as local as possible. For a local gauge transformation, we choose f to be a Kronecker-delta function:

$$f(r) = -f_0 \delta_{r,r_0} \tag{9}$$

localized at a lattice vertex r_0 as shown in Fig. 1, hence the discrete gradient becomes $\nabla f \simeq 0 - (-f_0) = f_0$ (the sign is arbitrary), and only the A operators emanating from this site are affected by the transformation:

$$A_{\langle r_0 r_1 \rangle} \to A_{\langle r_0 r_1 \rangle} + f_0, \tag{10}$$

where r_1 are nearest neighbor sites to r_0, and the edge direction matters. For each link $\langle r_0, r_1 \rangle$ in Eq. (10), the gauge transformation is realized by

$$e^{if_0 E} A e^{-if_0 E} = A + f_0 \tag{11}$$

which can be checked by the Baker–Campbell–Hausdorff identity

$$e^X \, A \, e^{-X} = A + [X, A] + \frac{1}{2!}[X, [X, A]] + \cdots . \tag{12}$$

Repeating Eq. (11) for the for link connecting to r_0, the complete local gauge transformation is then represented by the operator

$$U_{r_0}(f_0) = \prod_{r_1} e^{if_0 E_{\langle r_0 r_1 \rangle}}, \tag{13}$$

which represents the gauge transformation centered on site r_0, as shown by the red links in Fig. 1. The operator A takes values in $[0, 2\pi)$, so f_0 also takes values in $[0, 2\pi)$, ensuring the operator is well-defined. Each local gauge transformation forms a group $U(1)$, making this a quantum $U(1)$ gauge theory.

For now we have shown the U(1) gauge transform on a lattice, but we have not talked about the gauge-invariance condition, which will be discussed in the next subsection.

1.2 Lattice gauge invariance and correspondence in TC

To make things explicit, we first consider the continuum limit, where the gauge transformation in Eq. (13) becomes

$$U_{r_0}(f_0) = e^{if_0 \int_{S_{r_0}} dS \cdot E}, \tag{14}$$

where S_{r_0} is a small surface surrounding r_0. The invariance of a state $|\psi\rangle$ under these gauge transformations is defined upto a phase factor:

$$U_{r_0}(f_0) \, |\psi\rangle = e^{i\theta} \, |\psi\rangle \tag{15}$$

where the phase $e^{i\theta}$ encodes the the charge of the gauge field. For example, for a state whose phase after transformation is $\theta = 0$, the gauge invariance is equivalent to imposing Gauss's law

$$\int_{S_{r_0}} dS \cdot E = 0 \quad \Rightarrow \quad \nabla \cdot E = 0 \tag{16}$$

Now consider the case where the $U(1)$ group is broken down to Z_2. The operator A and E thus take values:

$$A \in \{0, \pi\}, \quad E \in \{-1, +1\} \tag{17}$$

To see its correspondence to Pauli matrices, it is convenient to rewrite them into:

$$e^{iA} \in \{-1, +1\}, \quad e^{i\pi E} \in \{-1, +1\} \tag{18}$$

Using the Baker-Campbell-Hausdorff identity, it is straightforward to check their anti-commutation relation:

$$e^{i\pi E}e^{iA}e^{-i\pi E}=e^{i(A+\pi)}=-e^{iA}\quad\Rightarrow\quad e^{i\pi E}e^{iA}=-e^{iA}e^{i\pi E}\quad\Rightarrow\quad\{e^{iA},e^{i\pi E}\}=0 \tag{19}$$

Recall that both e^{iA} and $e^{i\pi E}$ square to identity, thus the gauge field degrees of freedom reduce to a qubit represented by Pauli matrices by following identification:

$$\sigma^z\equiv e^{iA},\quad \sigma^x\equiv e^{i\pi E},\quad \sigma^y\equiv ie^{i\pi E}e^{iA} \tag{20}$$

It is readily to check that they obey the Pauli-matrix algebra:

$$[\sigma^z,\sigma^x]=2i\sigma^y,\quad [\sigma^x,\sigma^y]=2i\sigma^z,\quad [\sigma^y,\sigma^z]=2i\sigma^x,\quad \{\sigma^\alpha,\sigma^\beta\}=2\delta_{\alpha\beta} \tag{21}$$

We can now explain why the Toric Code is a Z_2 gauge theory. Using the correspondence between $e^{iA},e^{i\pi E}$ and σ^z,σ^x, the gauge transformation defined in Eq. (13) can be written as:

$$U_{r_0}=\prod_{j\in+}e^{i\pi E_j}=\prod_{j\in+}\sigma_j^x\equiv A_+ \tag{22}$$

with r_0 given by the central vetex shared by four edges (the center of the $+$). A gauge-invariant state under this gauge transform satisfies

$$A_+\left|\psi\right\rangle=e^{i\theta}\left|\psi\right\rangle \tag{23}$$

with $\theta=0$ corresponding to the sourceless electric Gauss law $\nabla\cdot E=0$ [in keeping with Eq. (14)]. Furthermore, if $\theta=\pi$, we have effectively

$$A_+\left|\psi\right\rangle=-\left|\psi\right\rangle\quad\Rightarrow\quad\int_{S_{r_0}}dS\cdot E=\pi \tag{24}$$

meaning there is a π electric charge at the vertex ($\nabla\cdot E=\pi$). This is why an excitation $A_+=-1$ in TC is often referred to a an electric charge e.

Similarly, we can also define the gauge-invariance condition for the magnetic flux, i.e.

$$e^{i\int dl\,A_l}\mapsto e^{i\sum_{j\in\Box}A_j}=\prod_{j\in\Box}\sigma_j^z\equiv B_\Box \tag{25}$$

Gauge-invarant magnetic fluxes requires

$$B_\Box\left|\psi\right\rangle=e^{i\alpha}\left|\psi\right\rangle \tag{26}$$

for $\alpha=0$, we have the zero-flux condition $B_\Box\left|\psi\right\rangle=\psi$, $\sum_{j\in\Box}A_j=0$; for $\alpha=\pi$, we have $B_\Box\left|\psi\right\rangle=-\left|\psi\right\rangle$, and a magnetic π flux given by $\sum_{j\in\Box}A_j=\pi$. This is why a state with with $B_\Box=-1$ in TC is often referred to as a magnetic flux excitation.

To energetically enforce a zero-electic-charge and zero-magnetic-flux state, we arrive at the TC Hamiltonain as illustrated in Fig. 2.

$$H=-\sum_+A_+-\sum_\Box B_\Box \tag{27}$$

with the commutation between A_+ and $B_\Box$ ensures that the gauge invariance for both electic and magnetic field can be satisfied simultaneously.

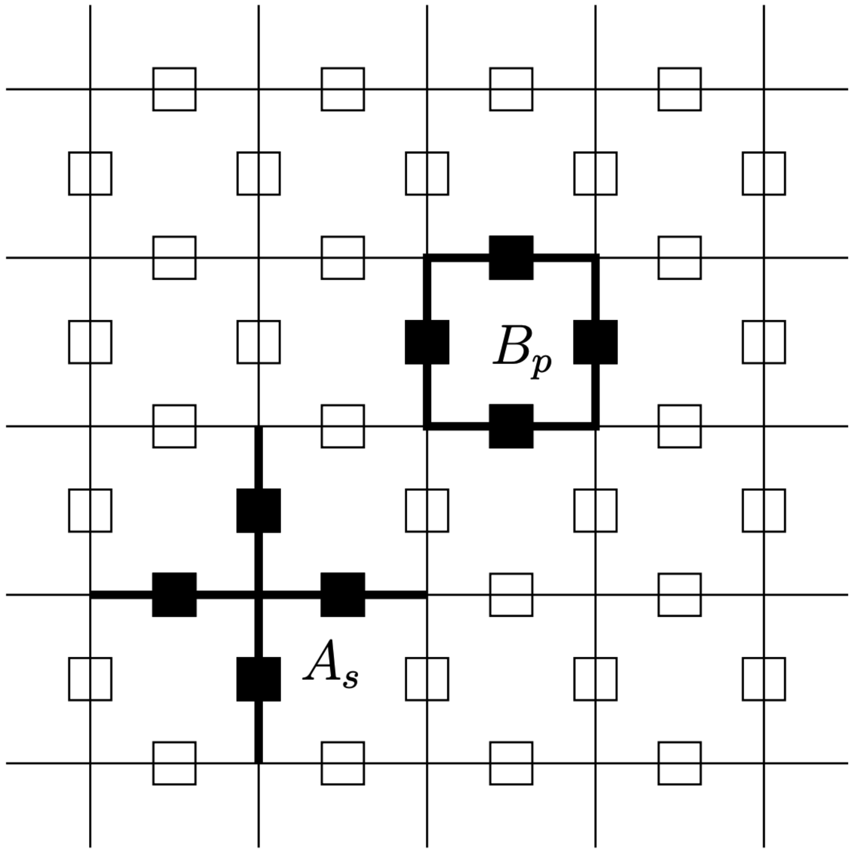

Figure 2: Toric code lattice $A_s \equiv A_+$, $B_p \equiv B_\square$.

Note that in usual electromagnetism, gauge symmetry is strictly enforced, meaning non-gauge invariant quantities are considered unphysical and not allowed. In the case of the TC, gauge symmetry is *imposed energetically*. As long as the system's energy is low enough (so as to stay in the ground state), or remain in an eigen state (in presence of non-zero electic charge or magnetic flux), the symmetry is not violated. However, generic high-energy states in the total Hilbert space may violate the gauge symmetry, for example, we can easily write down a high-energy product state that breaks the gauge symmetry in TC:

$$A_+ \left|\uparrow\uparrow\uparrow\uparrow\right\rangle = \left|\downarrow\downarrow\downarrow\downarrow\right\rangle \tag{28}$$

which obviously violates the gauge-invariance condition given by Eq. (23). This is a trade-off for formulating gauge theory using local, independent degrees of freedom.

2 Toric Code by Gauging an Ising Paramagnet

In the transverse-field Ising model, the Hamiltonian at the paramagnetic limit (no spontaneous symmetry breaking) is

$$H \;=\; -\sum_v \sigma_v^x \,, \tag{29}$$

where σ_v^x acts on the spin-$\frac{1}{2}$ degree of freedom at vertex site v. This model has a global $\mathbb{Z}_2$ symmetry

$$U \;=\; \prod_v \sigma_v^x \tag{30}$$

and the unique, gapped ground state is

$$|\psi\rangle \;=\; \bigotimes_v \frac{|\uparrow\rangle_v + |\downarrow\rangle_v}{\sqrt{2}}\,. \tag{31}$$

Our goal is to "gauge" this global $\mathbb{Z}_2$ symmetry, promoting it to a local symmetry. We introduce a $\mathbb{Z}_2$ gauge field (a qubit) on each edge e, with Pauli operators τ_e^x, τ_e^z. We define the local gauge transformation at vertex v by

$$U_v \;=\; \sigma_v^x \prod_{e\ni v} \tau_e^x\,, \tag{32}$$

where we used subscript convention commonly used in the context of gauging: $e \ni v$ denotes edges that share the same vertex v (similarly, $e \in p$ denotes edges that belong to a plaquette p). This enforces a $\mathbb{Z}_2$ Gauss law at v:

$$\sum_{e\ni v} E_e \;=\; n_v \pmod 2\,,$$

where n_v is the $\mathbb{Z}_2$ charge carried by the σ-spin.

To build a gauge-invariant Hamiltonian which include both matter and gauge, we:

1. Keep the transverse-field term (the matter part), since it already commutes with all U_v: $-\sum_v \sigma_v^x$.
2. Enforce gauge symmetry energetically by including the vertex term $-\sum_v U_v$.
3. Enforce zero flux on each plaquette p by adding $F_p \;=\; \prod_{e\in p} \tau_e^z$, and including $-\sum_p F_p$.

Thus the fully gauged Hamiltonian is

$$H_g \;=\; -\sum_v \sigma_v^x \;-\; \sum_v U_v \;-\; \sum_p F_p \;=\; -\sum_v \sigma_v^x \;-\; \sum_v \big(\sigma_v^x \prod_{e\ni v} \tau_e^x\big) \;-\; \sum_p \prod_{e\in p} \tau_e^z\,. \tag{33}$$

If we enforce $U_v = +1$ as a *hard* constraint (or take its coupling to $+\infty$), we may drop the σ-spins entirely (since at the ground state σ is completely determined by its surrounding gauge fields) and simplify to a pure gauge theory

$$H_{\mathrm{TC}} \;=\; -\sum_v \prod_{e\ni v} \tau_e^x \;-\; \sum_p \prod_{e\in p} \tau_e^z\,, \tag{34}$$

which is precisely the Toric Code Hamiltonian.

Remarks:

- Gauge-electic-charge (e) excitations are created by strings of the form $\sigma_v^z\, \tau^z_{\langle vu\rangle}\, \tau^z_{\langle uw\rangle} \cdots \sigma_w^z$, which commute with all U_v along their length.
- Gauge-magnetic-flux (m) excitations are created by strings of τ_e^x operators and likewise commute with all F_p except at their ends.
- In the simplified theory (34), the e and m excitations exhibit the characteristic mutual π-statistical phase of the $\mathbb{Z}_2$ gauge theory.

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